

cannon of the Napoleonic era. However, from the 1870s, breech-loading, rifled artillery, pioneered by the Prussians, transformed both range and rate of fire. Firing high-explosive steel shells, the French 75 field gun, introduced in 1898, could fire up to 30 rounds a minute up to 8.5km (5 miles).

LOGISTICS AND COMMUNICATION

Military commanders were mostly quick to adopt new technology. Two of the most important influences on 19th-century warfare – railways and the electric telegraph – were civilian inventions. Large-scale military use of railways came in 1859, when the French moved over 100,000 soldiers by train to fight the Austrians in northern Italy; soon after, both the railway and the telegraph were to prove invaluable in the American Civil War.



▲ LONG-RANGE SIEGE GUNS

During the Franco-Prussian War of 1870–71, Paris was bombarded with heavy Krupp siege guns. The superior performance of these breech-loading cannon was a major factor in Prussia's victory in the war.

◀ CIVIL WAR CAVALRY

While cavalry still had an important role in the American Civil War, as in conflicts of old, the traditional charge with sabres drawn was increasingly rare, as the revolver and carbine proved their superiority.

KEY EVENTS

19TH CENTURY

■ **1854–56** The rifled musket is employed in the Crimean War, which also sees use of the electric telegraph, and the beginnings of war photography.

■ **1863** At the Battle of Gettysburg, an assault by Confederate infantry – known as Pickett's Charge – fails, with 50 per cent casualties caused by the combined firepower of Union rifled muskets and cannon.

■ **1866** Prussia defeats Austria in the Seven Weeks' War, helped by its efficiency in mobilizing its forces by rail, and its use of the breech-loading Dreyse needle gun.

■ **1870–71** France is defeated in the Franco-Prussian War, chiefly because of the superiority of Prussian artillery.

